GROMM v1-4,3

Wider-grid calibration and VIX
Addendum to *Generative Risk-neutral Option Market Models*

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This addendum reports only the new wider-grid experiment and its comparison with the original GROMM v1. The main paper, `main2.tex`, is unchanged and contains the model theory, proofs and earlier experiments. Here we record the new calibration, fresh risk-neutral diagnostics and historical VIX comparison; implementation details and reproducibility information are collected in Appendix A.

1 A wider-grid GROMM v1 calibration and VIX

We consider a separately calibrated extension of the original GROMM v1, `Gromm-v1-4,3`, intended to improve coverage of the option wings relevant to variance swaps and VIX. It has the same SDLV architecture: 25 daily expiries, 200 interior strikes, 5,000 direct state coordinates per date, and an independent 5% Black smoothing factor. Both the 100-node one-day template and the 100-node 25-day template are widened to NMS $[-4, 3]$; the one-day grid is retained. NMS denotes the main paper's regularized, ATM-volatility-scaled log-moneyness coordinate; the original templates span $[-2.5, 1.5]$. The templates are merged into a common support as before. This changes the strike coverage, not the model's 25-business-day maturity limit. With the interior-node count held fixed, the wider support redistributes strike resolution rather than adding state coordinates.

1.1 Fresh calibration and market fit

The new fit uses the same 891 dates, 3 January 2022–29 August 2025, and the same upstream market-quality filters, but removes the final NMS quote cutoff. All 3,141,110 remaining call and put intervals are used. Quotes with their own expiry-specific NMS in $[-3.5, 2.5]$ receive weight one; all others receive weight 0.1. This is not a claim to retain every raw exchange quote. Each day's weighted loss is normalized by its sum of quote weights, and the objective averages equally across days. The loss remains bid–ask excess plus 0.05 times absolute midpoint error, in half-spread units, together with the original DLV cap penalty.

All daily model states were fitted from an ATM initializer, without transferring the original trained weights. The complete original schedule was run: 300 Adam epochs with cosine learning rate $0.02 \to 0.001$, followed by 700 epochs with $0.005 \to 0.00005$, resetting Adam at the continuation boundary. No VIX observations entered this objective.

Fit / quote subset	Quotes	% excess	Mean excess	Cond. mean
Original v1, original range	2,192,458	2.679%	0.03450	1.28797
Wider fit, original range	2,192,458	2.636%	0.03435	1.30311
Wider fit, core	2,465,684	2.701%	0.03484	1.28996
Wider fit, wings	675,426	31.635%	1.31125	4.14489
Wider fit, all quotes	3,141,110	8.923%	0.30930	3.46650
Wider fit, 29 Aug. (all)	4,190	15.012%	0.62411	4.15740

Table 1: Static-fit errors beyond the bid–ask interval, in half-spread units. All statistics are unweighted quote-level summaries, not the weighted training objective. Conditional means include only quotes with positive excess. The original-range rows cover the same 2,192,458 retained quotes; the final row is a separate single-date summary.

Table 1 separates the expanded quote universe from the original comparison set. The wider fit covers 97.299% of the core quotes, whereas the downweighted wings remain appreciably harder to fit. Across the full expanded universe, the conditional median excess is 1.98624 half-spreads and midpoint MAE is 0.56549 half-spreads. The original-range rows should be used when comparing static fit with the original GROMM v1; the all-quote and single-date rows have different denominators. Because grid coverage, quote scope and loss weights change together, this experiment does not isolate the effect of grid width alone.

1.2 Fresh diffusion estimation and risk-neutral audit

The diffusion is re-estimated from the new historical fitted surfaces, not transferred from GROMM v1. The estimator uses 710 exact one-session pairs ending no later than 31 December 2024, with the same fixed-contract alignment and Gaussian smoothing widths of one expiry and two strike nodes. The eight-factor system, including the front-spot factor, captures 75.87% of the smoothed innovation variance. Thus the 29 August 2025 initial state is contemporaneously calibrated, while the diffusion-estimation window ends before that date. Appendix B shows the new mean implied-volatility surface, the saved innovation spectrum and all eight retained factor responses.

We use the same front-density bridge and recomputed risk-neutral DLV drift as in the original model, with eight uniform substeps per day. From the completed new state we generate 100,000 fresh paths arranged as 50,000 independent antithetic pairs. The 20-business-day trading universe is spot plus 44 options, constructed from the new initial surface with the same NMS and maturity rule as the main paper: eleven initially OTM calls or puts at each of 5, 10, 15 and 20 business days, with cash strikes fixed thereafter. Appendix A.2 defines the reported statistics. Both conditional policies are trained afresh, with whole paths and complete pairs kept together in each fold. No old diffusion factors, simulated populations or trained conditional policies are reused.

N	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$N_{2.576}$	N_5	T_{sqr}	T_{ent}
570	2.426	2.118	0	0	0.995	0.995

Table 2: Fresh GROMM v1-4,3 restricted-universe $\mathbb{Q}$ audit: 100,000 paths, 50,000 antithetic pairs, 20 business days, spot plus 44 options; eight factors and eight steps per day. The statistics and colour conventions are those of Appendix A.2.

None of the 570 day–instrument coordinates exceeds 2.576 standard errors. The squared policy's out-of-fold mean and pair-based standard error are $(1.311, 1.317) \times 10^{-9}$ daily spot units,

with 95% interval $[-1.270, 3.892] \times 10^{-9}$. For the entropy policy the corresponding values are $(1.312, 1.318) \times 10^{-9}$ and $[-1.272, 3.896] \times 10^{-9}$. Both intervals contain zero. These diagnostics are consistent with the risk-neutral null for the tested instrument universe and policy class; they are not a proof over all strategies. In particular, the wider model has *not* undergone the main paper's 25-day full-instrument-grid audit, nor a separate test of every new wing strike. No new physical-measure simulation audit is claimed here.

1.3 Historical VIX comparison

Figure 1 compares the original and wider models with the official Cboe closing VIX and a reconstruction from the filtered market spreads supplied to SANOS. The discrete model series reprice the same market-selected option strips, with observed bid cutoffs and the same expiry interpolation. The analytic series instead use the model's full-strike log-contract variance at those same two expiries. Appendix A specifies the formulas and clocks.

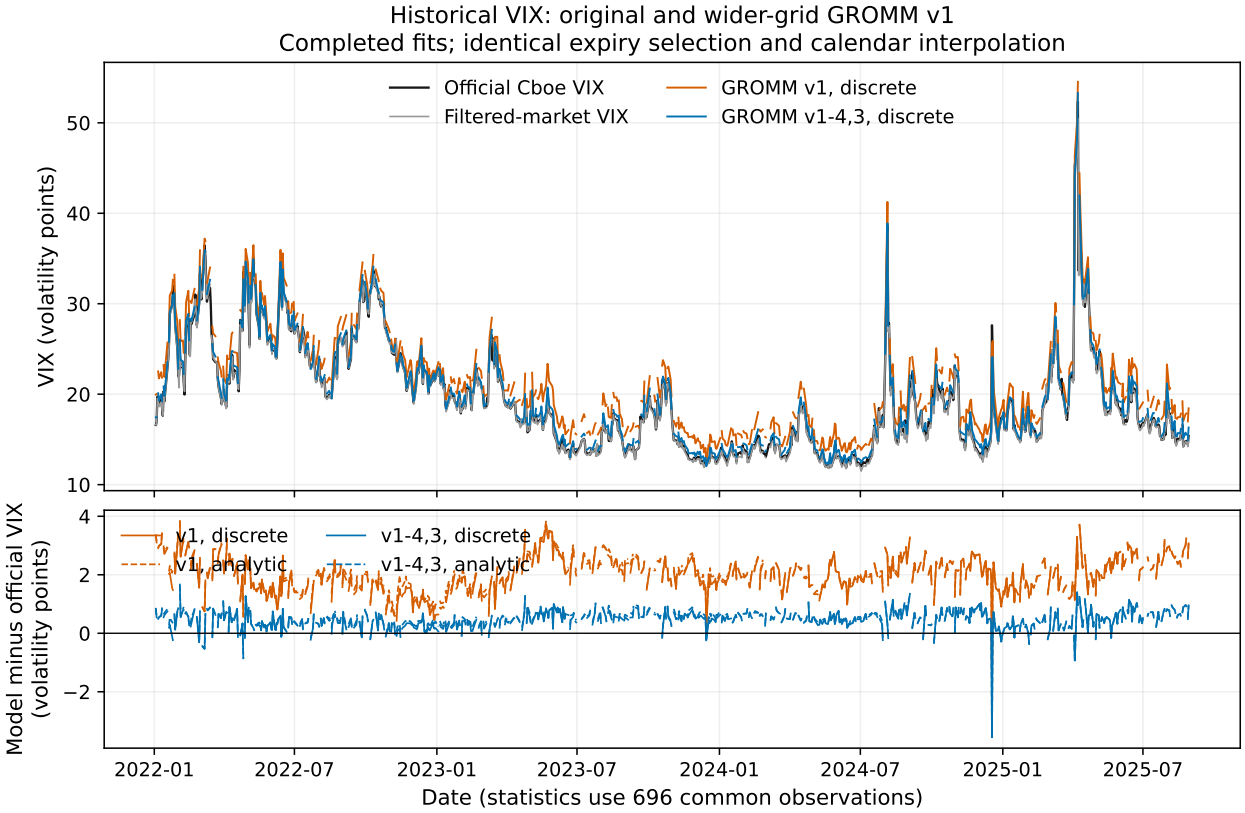


Figure 1: Historical VIX levels (top) and model errors relative to the official close (bottom). Orange denotes the original GROMM v1 and blue the completed GROMM v1-4,3 calibration. Solid model curves use the market-selected discrete strip; dashed error curves use the analytic full-strike variance swap. The grey top curve uses filtered market input spreads, not fitted SANOS prices. Statistics use the same 696 complete dates; missing model maturities are not extrapolated or filled in.

Series	Bias	MAE	RMSE	Corr.	29 Aug.
Raw-market strip	+0.028	0.142	0.265	0.99903	15.405
Filtered SANOS input	-0.181	0.235	0.349	0.99873	15.197
GROMM v1, discrete	+1.999	2.005	2.094	0.99446	18.434
GROMM v1, analytic	+2.062	2.068	2.149	0.99481	18.432
GROMM v1-4,3, discrete	+0.461	0.495	0.568	0.99848	16.314
GROMM v1-4,3, analytic	+0.497	0.528	0.595	0.99853	16.318

Table 3: Historical VIX comparison on the same 696 complete dates. Errors and the reference-date levels are in volatility percentage points; the benchmark is the official Cboe close (15.360 on 29 August 2025). Analytic values use the full-strike log contract at the same two expiries and the same calendar interpolation as the discrete strip, not direct evaluation at a different 30-day business maturity.

Discrete VIX MAE falls from 2.005 to 0.495 volatility points, a 75.33% reduction on identical dates. The wider model's analytic series has MAE 0.528 and RMSE 0.595; its remaining positive bias shows that the wider calibration does not fully remove the discrepancy. On the reference date the official close is 15.360, compared with 18.434 for the original discrete model and 16.314 for the wider discrete model.

There are 696 common model observations out of 891 fitted dates: on 186 dates a required Friday maturity exceeds the 25-business-day horizon, and nine dates lack an eligible expiry bracket. The raw-market and filtered-input reconstructions have 882 observations individually; Table 3 restricts them to the same 696 dates as the models. This is a historical comparison of contemporaneously calibrated surfaces, not an out-of-sample VIX forecast or a fit to VIX. Although the implemented discrete formula follows the Cboe strike-selection and interpolation rules, archived OptionMetrics end-of-day quotes and zero curves do not reproduce Cboe's synchronous inputs and intraday filtering. We therefore do not claim an exact reconstruction of the published closing tick.

A Wider-grid calibration and VIX implementation

A.1 State representation and checks

The wider model has 200 interior strikes plus two boundary nodes; the boundary levels remain one half of the smallest interior strike and one and a half times the largest. All 25 expiries share that support on a given historical date. The support floats between historical dates but cash strikes remain fixed within a simulated path and within each historical one-session alignment. The DLV-squared parameterization retains minimum 10^{-4} and upper cap 9, with the original log-space barrier. The completed fit's largest DLV-squared value is 6.319, with no coordinate at the cap. Across all historical states, independent float64 reconstruction has minimum probability 3.46×10^{-24} and maximum mass/forward errors below 4.3×10^{-14}.

Training uses float32 as in the baseline. Its finite-difference point-mass initializer has a small signed boundary roundoff of about -1.79×10^{-7}. Such checkpoints are accepted only when negative mass is bounded by that initializer's roundoff and independent float64 reconstruction passes strict positivity, unit-mass and unit-forward checks. The Q simulation and its initial-state reconstruction use float64. This numerical calibration check is separate from the dynamic drift tests.

A.2 Diffusion and audit conventions

Historical log-DLV changes are aligned at fixed expiry and cash strike; the endpoint surface is interpolated at origin relative strike divided by the realized gross spot return. A nodewise physical AR fit removes the conditional mean before smoothing and the joint spot/surface PCA. For the Q audit the drift is recalculated from the density recursion, not obtained by setting the empirical physical drift to zero. The simulator retains the paper's log-Euler step and its exceptional relative-drift taming threshold 0.5; it is not an exact finite-step martingale correction. The independent 5% Black factor has zero Q drift. The two conditional objectives, fold construction and spot normalization are those of the main paper; each fold uses a 30-epoch affine warm start and 30-epoch affine-plus-deep fit, with two width-32 SiLU layers, learning rate 10^{-3} and batch size 2,048. The simulation seed is 20261024. Constant physical risk prices are also re-estimated, but the new reported path diagnostics are under Q only.

For each live day–instrument coordinate, the return is $R^i_{p,n} = (V^i_{p,n+1} - V^i_{p,n})/Z_{p,n}$, where Z is pre-step spot and V^i is spot or an option price. The unconditional score is its sample mean divided by the standard error computed across 50,000 antithetic-pair averages. Table 2 reports the maximum and empirical 99th percentile of $|z|$ over the 570 coordinates, together with counts N_c for $|z| > c$. These coordinatewise thresholds are not family-wise p-values.

Conditional policies use current Brownian-factor histories, log spot, live log option-price-to-spot ratios, time and remaining maturities. For an action $a(X)$, write $r = a(X)^\top R$. The squared objective maximizes the training mean of $r - \frac{1}{2}\|a\|^2 - \frac{1}{2}r^2$; the entropy objective minimizes $\log \text{mean} \exp(-r + \frac{1}{2}\|a\|^2)$. Complete antithetic pairs are split into two equal folds; input standardization and action fitting use only the training fold, and each frozen policy is evaluated on the other fold. The folds are then exchanged. For either objective, the reported T is the mean out-of-fold daily portfolio return divided by its pair-based standard error, after averaging over the 20 days within each path. The main paper's per-test colour rule is black for $|T| \le 1.96$, orange for $1.96 < |T| \le 2.576$ and red above 2.576. Both new conditional statistics are therefore black.

A.3 Market-selected discrete variance

For each eligible expiry let D_T be the discount factor, F_T the put–call-parity forward, K_0 the greatest available strike not exceeding F_T, and $O_T(K)$ the selected OTM midpoint price (the average of call and put prices at K_0). The Cboe total-variance formula is

$$v_T^{\text{strip}} = \frac{2}{D_T} \sum_i \frac{\Delta K_i}{K_i^2} O_T(K_i) - \left(\frac{F_T}{K_0} - 1 \right)^2 . \tag{1}$$

Interior ΔK_i are half the neighbouring-strike difference; endpoints use the adjacent-strike difference. Selection walks outward from K_0, omits zero-bid contracts and stops after two consecutive zero bids on each side.

Friday expiries strictly between 23 and 37 calendar days are selected with their AM/PM settlement convention. For minute counts $N_1 < N_{30} < N_2$,

$$\mathrm{VIX} = 100\sqrt{\frac{N_{365}}{N_{30}}\left[\frac{N_2 - N_{30}}{N_2 - N_1}v_{T_1} + \frac{N_{30} - N_1}{N_2 - N_1}v_{T_2}\right]}, \tag{2}$$

where $N_{30} = 43,200$ and $N_{365} = 525,600$. These conventions follow the Cboe VIX methodology, pp. 2–7.

The model replaces only the option prices in (1); the observed raw bid cutoffs, forward, K_0, discount factors, contracts and interpolation times are shared by both GROMM variants. No fictitious model bid/ask quotes are introduced. Filtered-market VIX uses the synthetic call-equivalent bid/ask midpoints passed to SANOS, including parity conversion for puts, not fitted option prices. Since preprocessing has removed zero bids, the filtered-strip calculation respects the observed raw-book cutoffs, then computes its own forward, K_0 and strike quadrature on the surviving input quotes.

A.4 Analytic full-strike variance

For the forward-normalized mixture $S_T/F_T = s_J B_T$, with probabilities $p_j(T)$ on positive support s_j and independent mean-one Black factor of business-time variance $0.05^2\tau_b(T)$, the model gives

$$v_T^{\text{analytic}} = -2\mathbb{E}^{\mathbb{Q}}[\log(S_T/F_T)] = -2\sum_j p_j(T)\log s_j + 0.05^2\tau_b(T). \tag{3}$$

This is the full-strike log-contract/variance-swap quantity, not a discrete VIX strip. We substitute these two expiry values into (2); a direct 30-calendar-day model evaluation has a different maturity convention and is not used in Table 3 or Figure 1.

Calendar time is measured from 16:15 New York to 09:30 for AM settlement or 16:00 for PM settlement. Model business time uses the NYSE session calendar divided by 252 and the implementation's 0.8-day AM subtraction. Fractional model maturities interpolate adjacent daily densities. Discount factors use the cached OptionMetrics zero curve with linear rate interpolation, not Cboe's Treasury spline. No maturity beyond 25 business days is extrapolated. These clock and input differences are retained in the reported comparisons, not calibrated away by a VIX rescaling.

A.5 Reproducibility

The fresh calibration is run by `quantdata/sdlv/daily/wide_fit.py`; the Q audit by `quantdata/sdlv/daily/wide_audit.py`; and the VIX calculations by `quantdata/sdlv/daily/vix_comparison.py` and `quantdata/dysanos/filtered_vix.py`, using the shared formula in `quantdata/vixdata/formula.py`. The source for the new tables and figure is `quantdata/sdlv/daily/paper_wide_grid.py`. The retained `generated/wide_grid/metrics.json` records the completed fit and audit, source paths and SHA-256 hashes; the accompanying `vix_series.csv` supplies the plotted observations. Large training data, checkpoints and simulation chunks remain outside the Git tree. This experiment uses training run `20261023-kVyePV` and Q-audit run `q-audit-20261024-4mkznc`. The VIX and Q results are checked to refer to the identical final 1,000-epoch checkpoint; no interim VIX snapshot is used in this addendum.

B Mean volatility surface and new PCA factors

B.1 Mean and reference-date implied volatility

Figure 2 shows the arithmetic mean of the wider model's Black implied-volatility surfaces over all 891 fitted dates, with equal weight per date. Each surface is evaluated on the same NMS $[-4, 3]$ and 1–25-business-day grid, using that date's market ATM volatility scale. This is a mean of implied volatilities, not the implied volatility of an averaged price surface, nor an average of the DLV parameters. The right panel is the 29 August 2025 fitted state used for the factor illustrations. Neither panel averages simulated paths. Both include the 5% Black smoothing component and use business-time annualization.

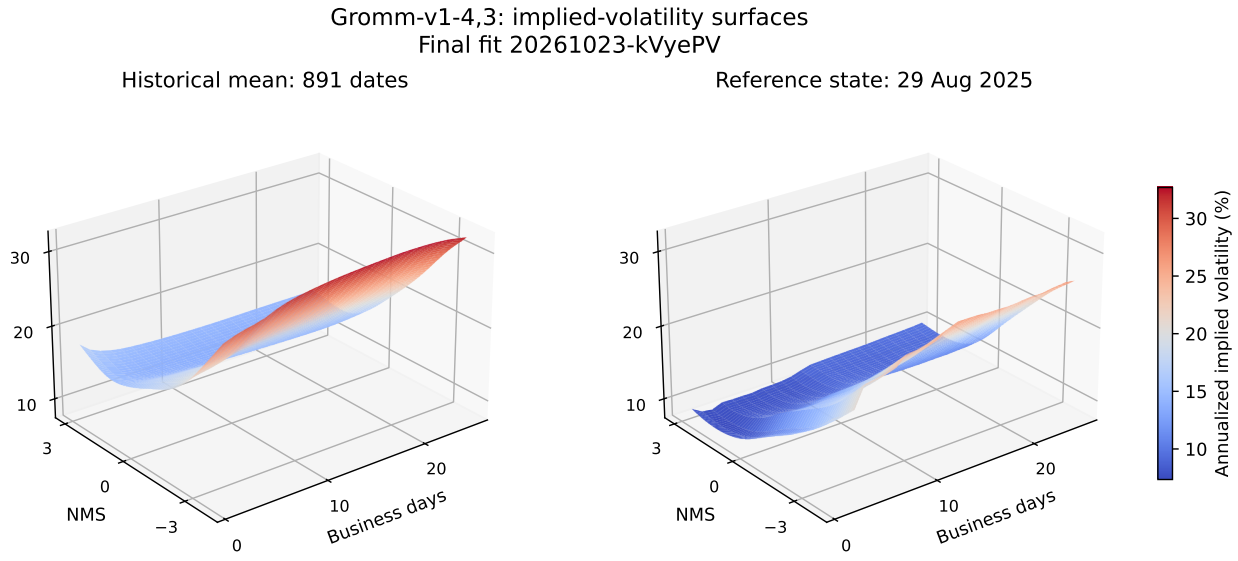


Figure 2: GROMM v1-4,3 annualized implied volatility in percent: historical equal-date mean (left) and reference-date state (right). Both panels use identical vertical and colour scales. Date-specific NMS alignment prevents the historical mean from mixing different standardized strikes. These are model surfaces over the displayed grid, including downweighted wings, not a claim of observed quote coverage at every plotted point.

B.2 Innovation spectrum

The new factor system is the one saved by the completed wider-grid Q audit, estimated from 710 fixed-contract historical pairs ending on 31 December 2024. The first direction is aligned to the front-spot innovation; the remaining seven retained directions are PCs of the smoothed log-DLV innovations after removing their spot projection. These are not PCs of implied-volatility levels. Figure 3 reports the spot-loading power followed by the residual covariance eigenvalues, each divided by total smoothed innovation variance. The eight retained factors explain 75.87%; the first 90% crossing is rank 25. Eigenvalues beyond rank eight are shown for context; those additional directions were not used in the reported Q audit.

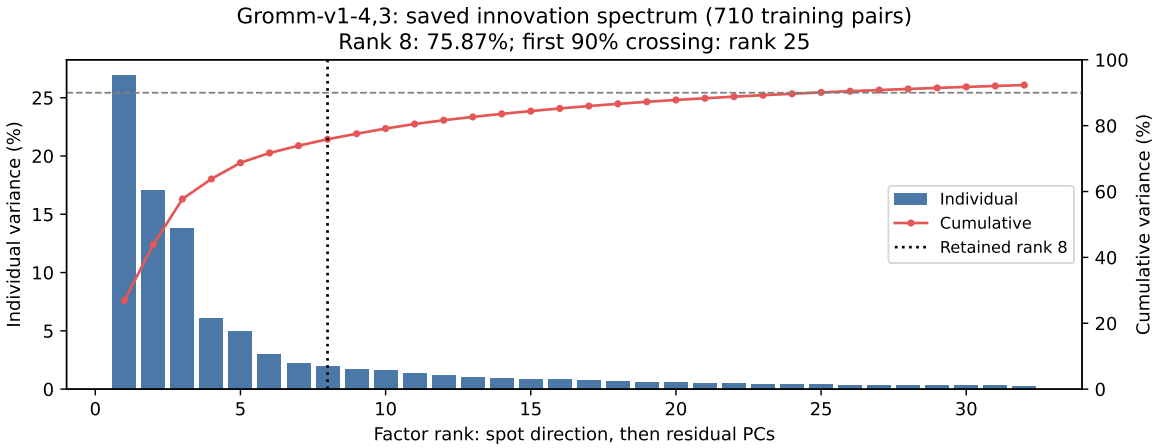


Figure 3: New spot-oriented factor spectrum, using the saved wider-grid diffusion estimate. Blue bars show individual variance contributions; the red line is cumulative. The vertical dotted line marks the eight retained factors, not the rank needed to reach the horizontal 90% line.

B.3 Implied-volatility responses of the retained factors

At the reference-date state, each saved annualized loading ξ_f is mapped to an implied-volatility response by the same central daily shock convention as the main paper:

$$\Delta_f \sigma_{\text{imp}} = \tfrac{1}{2}\left[\sigma_{\text{imp}}(\log \Sigma + \xi_f/\sqrt{252}) - \sigma_{\text{imp}}(\log \Sigma - \xi_f/\sqrt{252})\right]. \tag{4}$$

Only the future log-DLV-squared rows (days 2–25) are displaced. Spot, the first-expiry state, and the reference-date NMS-to-strike map are held fixed. Thus even for the spot-aligned factor this is its surface-loading component, not a full joint spot/front-density simulation. There is no drift or elapsed-time step in these illustrations; an equivalent measure change leaves the illustrated diffusion directions unchanged.

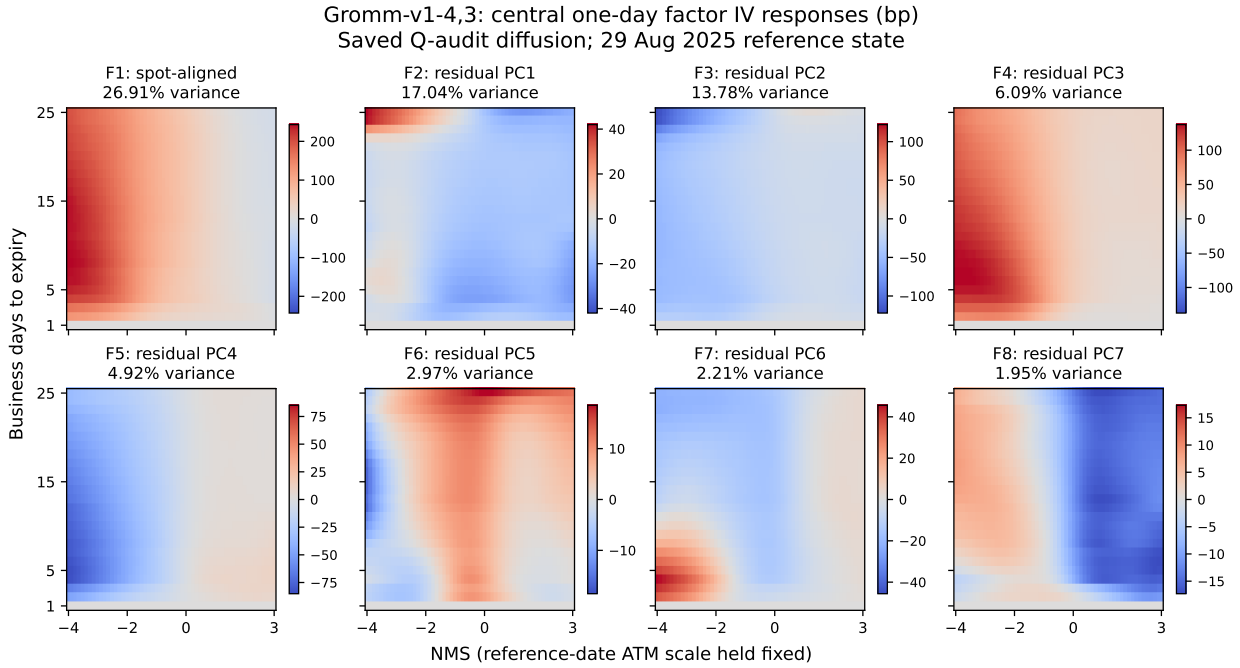


Figure 4: All eight new retained factors in IV coordinates. Each panel shows the central one-day-standard-deviation response in basis points of annualized IV (1 bp $= 10^{-4}$), using its own symmetric colour scale. Panel percentages are contributions to total smoothed log-DLV innovation variance, not IV variance. F1 is spot-aligned; F2–F8 are residual PCs 1–7. The one-day row is zero because its state is held fixed.

The figures are regenerated by `quantdata/sdlv/daily/paper_wide_factors.py`. Their plotted arrays and provenance are retained in `generated/wide_grid/factor_surfaces.npz` and `generated/wide_grid/factor_metrics.json`. Checkpoint identities, reference-state reconstruction and retained variance are verified against the same completed Q audit used in Table 2; no new calibration or path simulation is performed to make these illustrations.